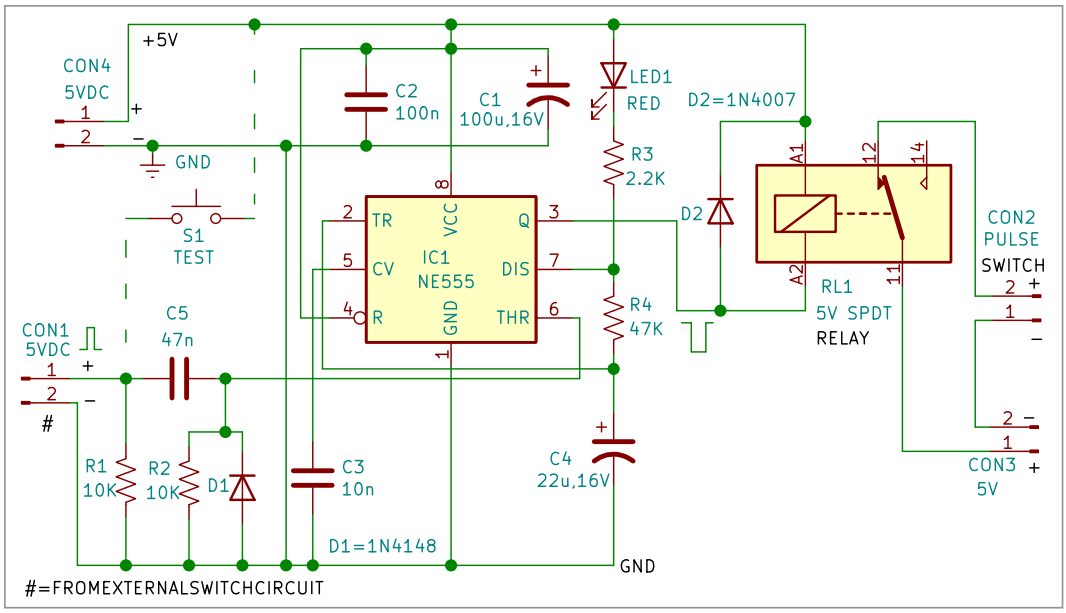


Fig. 2: Circuit diagram

BONUS CONTENT ONLINE!

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approximately 1.1 seconds, the timing cycle will complete. LED1 should turn off, and the relay should click a second time as it de-energises and drops back into its standby state.

Troubleshooting

If your circuit isn't behaving as expected, use this systematic checklist to diagnose and resolve the issue:

1. Relay chatter or random triggering.

• *Cause.* Often triggered by 5V supply dips, ripples, or electrical noise.

• *Fix.* Ensure your power supply is clean and stable. Verify that proper decoupling capacitors are connected close to the IC pins.

2. Incorrect or unstable relay ON-time.

• *Cause.* Faulty, incorrectly rated, or loose connections in the R4/C4 timing network, or electrical noise affecting the control-voltage (Pin 5) pin.

• *Fix.* Re-check the values and soldering of R4 and C4. If the timing varies randomly, add a small ceramic capacitor (typically 0.01µF) from Pin 5 to ground to filter out noise.

3. Immediate NE555 IC failure.

• *Cause.* A missing, faulty, or backward-installed flyback diode (D2) across the relay coil. Without it, inductive voltage spikes from the coil can instantly destroy the IC.

• *Fix.* Ensure D2 (e.g., 1N4007) is correctly connected in reverse-bias directly across the relay coil terminals.

4. External trigger failure.

• *Cause.* Component faults or bad solder joints within the input network consisting of CON1, C5, R1, R2, or D1.

• *Fix.* Trace the trigger pulse at Pin 2 using an oscilloscope or multimeter. Remember that the NE555 triggers on a LOW pulse; ensure that Pin 2 rests HIGH and briefly drops LOW only when the external trigger is applied.

General circuit check

• Double-check all component values, structural orientations (especially polarised capacitors and diodes), and look for cold solder joints.

• Verify that your grounding is solid across the entire PCB layout.

Applications

Use this circuit for control panel applications, or to drive momentary-input devices such as power buttons, gate triggers, and latching relays. EFY



T.K. Hareendran is an electronics designer, hardware beta tester, technical author, and product reviewer

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